

Product Laws of ExponentsQuestion Bank 3.01 **ANSWER KEY***Show your work. Circle your final answer.***For questions 1 and 2, write the product as a repeated multiplication expression, then as a single power.**

1. $3^3 \cdot 3^4$

$$(3 \cdot 3 \cdot 3)(3 \cdot 3 \cdot 3 \cdot 3) = 3^7$$

Seven factors of 3. Same base, add the exponents.

2. $5^1 \cdot 5^4 \cdot 5^3$

$$5 \cdot (5 \cdot 5 \cdot 5 \cdot 5)(5 \cdot 5 \cdot 5) = 5^8$$

The law extends to three powers: $1 + 4 + 3 = 8$.

3. Evaluate each power, then multiply. $4^1 \cdot 4^3$

$$4 \cdot 64 = 256$$

Also $4^4 = 256$, which is the check.

4. Apply the product of powers law, then evaluate the power. $(-8)^2 \cdot (-8)^6$

$$(-8)^8 = 16,777,216$$

Eight negative factors: positive. A calculator confirms 8^8 .

5. Jose says he can evaluate $9^3 \cdot 9^{12}$ using the product of powers law. Is he correct? Explain, and write the single power.

$$\text{Yes — the bases are the same, so } 9^3 \cdot 9^{12} = 9^{15}.$$

*The law needs matching bases and nothing else. (He would not need to evaluate 9^{15} to be right.)***For questions 6 through 8, use the power of a product law.**

6. Write $(4 \cdot 5)^2$ as a repeated multiplication expression, group the repeated factors, then write it in the form $a^m \cdot b^n$.

$$(4 \cdot 5)(4 \cdot 5) = (4 \cdot 4)(5 \cdot 5) = 4^2 \cdot 5^2$$

Value 400 either way.

7. Evaluate inside the parentheses first, then apply the exponent. $(-7 \cdot 5)^4$

$$(-35)^4 = 1,500,625$$

Even power of a negative: positive.

8. Apply the power of a product law, then evaluate. $(3 \cdot (-6))^2$

$$3^2 \cdot (-6)^2 = 9 \cdot 36 = 324$$

$$\text{Or } (-18)^2 = 324.$$

For questions 9 through 11, use the power of a power law or the identity law.

9. Write $[(3)^2]^8$ as a single power.

$$3^{16}$$

Multiply the exponents: $2 \cdot 8$.

10. Rewrite $(0.5 \cdot (-8))^2$ using a law of exponents, name the law, then evaluate.

$$0.5^2 \cdot (-8)^2 = 0.25 \cdot 64 = 16 \quad \text{— power of a product}$$

$$\text{Or } (-4)^2 = 16.$$

11. Write each expression using the identity law, then give its value.

a. 285.6^1

$$285.6$$

b. $\left(\frac{24}{8}\right)^1$

$$\frac{24}{8} = 3$$

Working backwards.

12. Find the value of n that makes each statement true.

a. $2^3 \cdot 2^n = 2^8$

$$n = 5$$

$$3 + n = 8.$$

b. $(5^2)^n = 5^{10}$

$$n = 5$$

$$2n = 10.$$

c. $(7 \cdot 3)^n = 7^4 \cdot 3^4$

$$n = 4$$

Power of a product read backwards.

13. Which expression is equivalent to $6^2 \cdot 6^5$?

- ☒ A. 6^7
- ☐ B. 6^{10}
- ☐ C. 36^7
- ☐ D. 12^7

A

Same base, add the exponents.

14. Select ALL of the expressions that are equivalent to 2^6 .

- ☒ A. $2^2 \cdot 2^4$
- ☒ B. $(2^3)^2$
- ☒ C. $2^3 \cdot 2^3$
- ☒ D. 4^3
- ☐ E. $2^2 \cdot 2^3$
- ☐ F. $(2^2)^4$

A, B, C and D

A: 2^6 . B: 2^6 . C: 2^6 . D: $4^3 = 64 = 2^6$ — same value, different base, still equivalent. E is 2^5 ; F is 2^8 .